20 An initially stationary firework explodes and splits into two fragments that move horizontally in opposite directions.

The total kinetic energy transferred to the fragments by the explosion is E.

One fragment has mass m and the other one has mass $2m$.

What is the speed of the fragment of mass m immediately after the explosion?

A $\sqrt{\frac{E}{m}}$ **B** $\sqrt{\frac{2E}{m}}$ **C** $\sqrt{\frac{2E}{3m}}$ **D** $\sqrt{\frac{4E}{3m}}$